Bellman Defect Correction

Inference Time Algorithms for SciML and GenAI

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More thought can improve the answer

• Flash

Quick response

YOU

A bat and a ball cost $1.10 together.
The bat costs $1.00 more than the ball.
What does the ball cost?

AI

The ball costs **$0.10**.
$1.10 - 1.00 = 0.10$

✗ Fails the check: $0.10 + $1.10 = $1.20

• Pro

Reason and verify

YOU

A bat and a ball cost $1.10 together.
The bat costs $1.00 more than the ball.
What does the ball cost?

AI

Let the ball cost x. The bat then costs $x + 1$.

$$x + (x + 1) = 1.10 \implies x = 0.05$$

The ball costs **$0.05**.

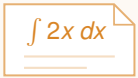
✓ Check: $0.05 + $1.05 = $1.10

Same question. More computation for reasoning and checks.

Illustrative AI dialogues

From fast answers to scientific discovery

Flash mode



Solve undergraduate problems

Reasoning



Develop and check research ideas

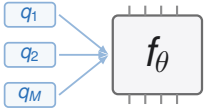
Large agent teams



Navier–Stokes blowup construction*

OUR SCALING SETUP

More queries.
Same model.



Fixed weights θ

More inference computation

Which algorithm turns more queries into better answers?

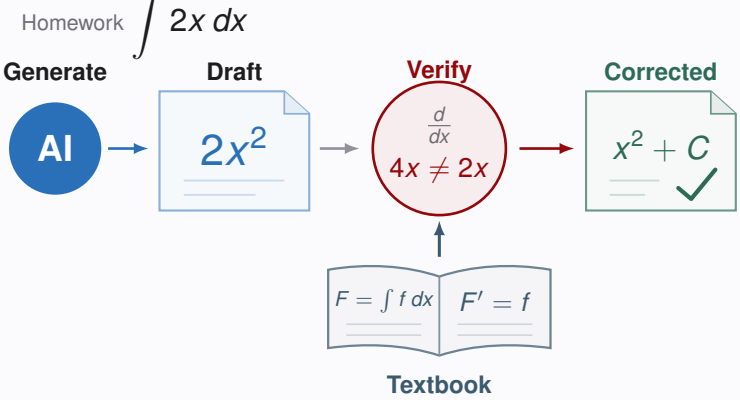
Illustrative task progression. *Reported NS case: ∼10,000 agents, with model upgrades. OpenAI (2026).

Textbook knowledge checks an AI answer

MY AI PHILOSOPHY

Generate first. Verify with known rules.

The error tells us what to correct.



Defect correction has exactly two steps

Write the Law of Defect

Derive the equation obeyed by $\check{u} := u - \widehat{u}$.

1

known equation + autodiff

Simulate the Law of Defect

Spend new compute on stochastic paths for $\check{u}$.

2

linear: Feynman Kac + Monte Carlo

semilinear: multilevel Picard

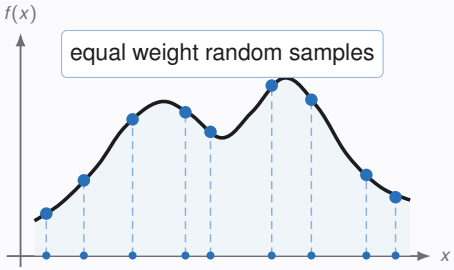
$$u_{\mathrm{corr}} = \widehat{u} + \widehat{\check{u}}_N$$

Write analytically. Spend inference compute stochastically.

No retraining: additional compute targets only what the model missed.

Source: SCaSML paper, structure preserving Law of Defect and stochastic correction.

Monte Carlo estimates an integral



random nodes estimate the area directly

SAMPLE, THEN AVERAGE

$$I(f) = \int f(x)\,\mu(dx).$$

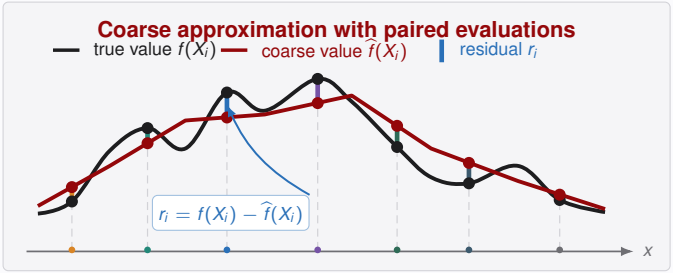
$$X_i \overset{\text{iid}}{\sim} \mu, \qquad \widehat{I}_N^{\text{MC}} = \frac{1}{N}\sum_{i=1}^{N} f(X_i).$$

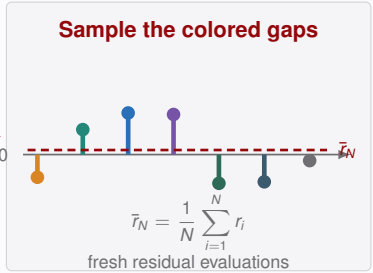
$$\mathbb{E}[\widehat{I}_N^{\text{MC}}] = I(f).$$

Sampling rate

$$\text{RMSE} = \frac{\sqrt{\text{Var}_\mu(f)}}{\sqrt{N}}.$$

Learn the bulk, simulate only the defect





$$I(f) = I(\widehat{f}) + \mathbb{E}[r(X)], \qquad \widehat{I}_{\mathrm{CV}} = I(\widehat{f}) + \bar{r}_N.$$

Training learns the bulk; inference averages only what the surrogate missed.

Residual Monte Carlo multiplies the two rates

Condition on a surrogate trained with independent data.

$$\mathbb{E}[\widehat{I}_{\mathrm{CV}} \mid \widehat{f}] = I(f), \qquad \mathrm{RMSE}(\widehat{I}_{\mathrm{CV}} \mid \widehat{f}) \le \frac{\|f - \widehat{f}\|_{L^2(\mu)}}{\sqrt{N}}.$$

Two independent gains multiply

$$\underbrace{\|f - \widehat{f}_m\|_2}_{O(m^{-\gamma})} \times \underbrace{N^{-1/2}}_{\text{residual samples}} = \underbrace{O(m^{-\gamma} N^{-1/2})}_{\text{corrected RMSE}}.$$

MATCHED BUDGETS: $N \asymp m$ $\qquad \mathrm{RMSE} = O(m^{-\gamma-1/2})$

Source: Blanchet, Chen, Lu, and Ying, arXiv:2305.16527, Theorems 4 and 5 for the noisy linear functional.

PDE correction uses the Bellman defect

LINEAR EQUATION

$$(\partial_t + \mathcal{L})u = f, \qquad u(T) = g.$$

For a learned $\widehat{u}$, set

$$\widehat{f} = (\partial_t + \mathcal{L})\widehat{u}, \qquad \delta_{\mathrm{B}} = f - \widehat{f}.$$

Subtract the two equations

$$\boxed{(\partial_t + \mathcal{L})\check{u} = \delta_{\mathrm{B}}.}$$

SEMILINEAR EQUATION

$$(\partial_t + \mathcal{L})u + F(u, z) = 0, \qquad z = \sigma^\top \nabla u.$$

For $\widehat{z} = \sigma^\top \nabla \widehat{u}$, define

$$\widehat{r} = (\partial_t + \mathcal{L})\widehat{u} + F(\widehat{u}, \widehat{z}).$$

The defect has its own equation

$$(\partial_t + \mathcal{L})\check{u} + \check{F}(\check{u}, \check{z}) = -\widehat{r},$$
$$\check{F}(v, w) = F(\widehat{u} + v, \widehat{z} + w) - F(\widehat{u}, \widehat{z}).$$

$$\check{u} = u - \widehat{u}, \qquad \check{z} = \sigma^\top \nabla \check{u}, \qquad \check{u}(T) = g - \widehat{u}(T).$$

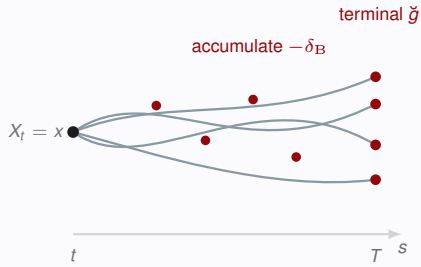
Keep the trained model and solve for its remaining error.

Source: SCaSML manuscript, linear and semilinear defect equations.

Feynman Kac simulates only the defect

SAMPLE THE GENERATOR

$$X_t = x, \qquad dX_s = \mu(s, X_s)\, ds + \sigma(s, X_s)\, dW_s.$$



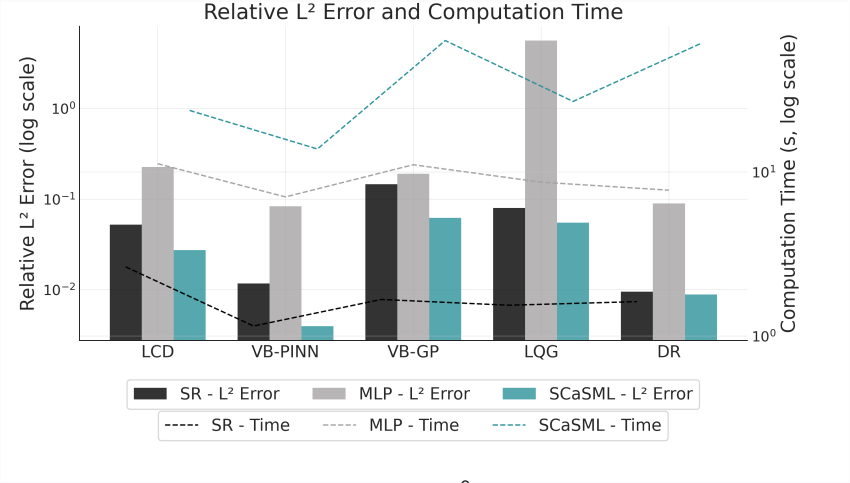
FEYNMAN KAC REPRESENTATION

$$\Delta := \breve{g}(X_T) - \int_t^T \delta_{\mathrm{B}}(s, X_s)\, ds$$
$$\breve{u}(t, x) = \mathbb{E}_{t,x}[\Delta].$$

MONTE CARLO REALIZATION

$$\widehat{u}^{(N)}_{\mathrm{corr}}(t, x) = \widehat{u}(t, x) + \frac{1}{N} \sum_{n=1}^{N} \Delta^{(n)}.$$

SCaSML spends inference compute to buy accuracy



More inference compute lowers relative L^2 error across all five benchmarks.

Diffusion sampling after training

A FIXED GENERATIVE MODEL

We write the sampling clock from noise at $t = 0$ to data at $t = T$.

$$X_0 \sim p_0, \qquad dX_t = b_t(X_t)\,dt + \sigma_t\,dB_t.$$

The drift b_t includes the pretrained score or denoising model.

One numerical transition

$$Y = x + hb_t(x) + \sqrt{h}\,\sigma_t Z, \quad Z \sim N(0, I).$$

A NEW TASK AT INFERENCE TIME

An observation, classifier, or critic assigns a terminal potential $\phi(x) > 0$.

Desired output distribution

$$\pi_T(dx) = \frac{\phi(x)p_T(dx)}{\mathcal{Z}},$$

$$\mathcal{Z} = \mathbb{E}[\phi(X_T)].$$

For reward r, take $\phi = e^{\beta r}$.

How should extra sampling compute improve this target match?

The value function guides noisy states

FUTURE REWARD GIVEN THE CURRENT STATE

$$H^*(t,x) = \mathbb{E}[\phi(X_T) \mid X_t = x].$$

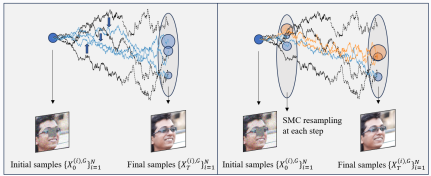
$$(\partial_t + \mathcal{L}_t)H^* = 0, \qquad H^*(T,x) = \phi(x).$$

Ideal Doob drift

$$b_t(x) + a_t \nabla \log H^*(t,x), \qquad a_t = \sigma_t \sigma_t^\top.$$

In practice, use a positive approximation $H \approx H^*$.

For random starts, the exact Doob law also tilts p_0 by $H^*(0,\cdot)$.



(a) Naïve Guidance (b) Unbiased Guidance via SMC

Reward scores a completed sample.

Value predicts its future reward.

Approximate guidance changes the sampling law. A correction must account for that change.

Source: URGE and Stein guidance manuscripts.

Path weights correct approximate guidance

A FIXED NUMERICAL GRID

$p_k(x, dy)$ is the base transition. A guided proposal uses $q_k(x, dy)$.

Exact importance weight for the discretized target

$$W = \phi(X_n) \prod_{k=0}^{n-1} \frac{p_k(X_k, X_{k+1})}{q_k(X_k, X_{k+1})}.$$

With $H_n = \phi$, the same weight factors into local corrections:

$$w_k = \frac{H_{k+1}(X_{k+1})}{H_k(X_k)} \frac{p_k(X_k, X_{k+1})}{q_k(X_k, X_{k+1})}, \qquad W = H_0(X_0) \prod_{k=0}^{n-1} w_k.$$

SMC implementation
Propagate, update weights, and resample when needed.

URGE viewpoint
Path likelihood ratios give the continuous Girsanov correction.

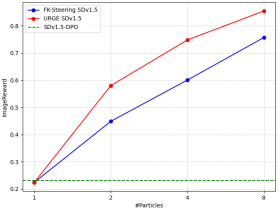
Source: URGE manuscript, path correction construction.

URGE for image restoration and reward guidance

FFHQ 256 SUPER RESOLUTION

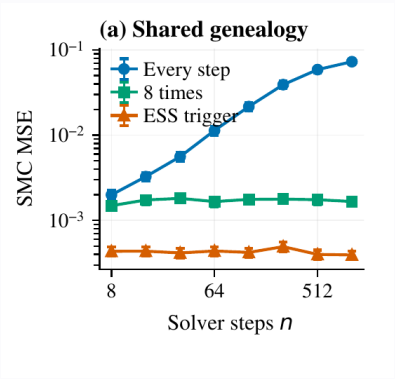


SD V1.5 PARTICLE SCALING



In the reported tests, URGE gives comparable restoration quality and higher ImageReward than FK Steering for $N = 2, 4, 8$.

SMC numerics under mesh refinement



SAME TARGET AND VALUE MODEL

Only the numerical mesh and resampling schedule change.

What the plot says

- Resampling at every solver step makes the MSE grow rapidly with depth n.
- A fixed number of resampling times or an ESS trigger remains nearly flat.

Repeated population interaction drives the observed instability.

Source: Stein guidance manuscript, numerical experiment, panel (a).

Perfect ESS can hide nonconvergence

Full multinomial resampling at every microstep; $X_0^{i,N} = 0$, $n, N \geq 2$.

NEUTRAL BROWNIAN CHAIN

$$X_p^{i,N} = X_{p-1}^{A_p^i,N} + n^{-1/2}\xi_p^i,$$

$$A_p^i \sim \mathrm{Unif}\{1,\dots,N\}, \qquad \xi_p^i \sim \mathcal{N}(0,1).$$

Every local check passes

$\phi \equiv 1$, $\mathrm{ESS}_p = N$, and $X_n^{i,N} \sim \mathcal{N}(0,1)$ exactly.

Yet the empirical mean does not stabilize

For $g(x) = \sin(x)/2$,

$$\mathbb{E}\left[\left(\frac{1}{N}\sum_{i=1}^{N} g(X_n^{i,N})\right)^2\right] \geq \frac{1}{32\mathrm{e}^2}\min\left\{\frac{n}{N}, 1\right\}.$$

For fixed N, the error stays nonzero as $n \to \infty$.

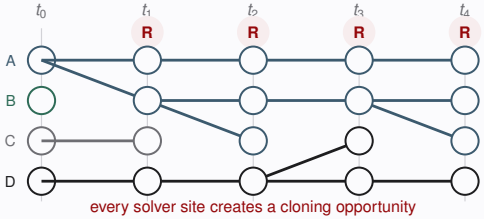
RMSE $\leq \varepsilon$ forces $N = \Omega(n\varepsilon^{-2})$ and work $= \Omega(n^2\varepsilon^{-2})$.

Scope: full multinomial resampling at every step. Complexity claims concern sufficiently small ε.

Source: Two Queries, No Gradients, Theorem 1 and Corollary 2. Full multinomial resampling at every step.

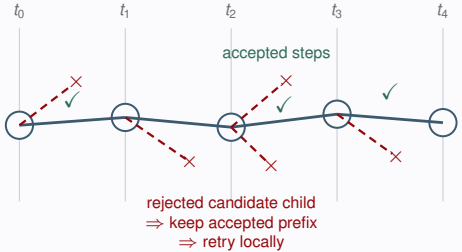
Sequential rejection preserves each trajectory

FULL MICRO RESAMPLING one parent pool



categorical ancestor draw from one shared particle pool

EXACT CHILD REJECTION independent proposal streams



one accepted trajectory with local accept/reject events

Resampling moves mass across lineages. Rejection spends serial trials within a lineage.

Approximate local values leave a path defect

THE LOCAL REJECTION DISTRIBUTION

$$G_k(x) = (P_k H_{k+1})(x),$$

$$Q_k^H(x, dy) = \frac{p_k(x, dy) H_{k+1}(y)}{G_k(x)}.$$

Rejection samples this transition without evaluating G_k, given a valid envelope.

Its output is an unweighted trajectory.

THE MISSING PROBABILITY COMPENSATION

For $H_n = \phi$

$$R_H = H_0(X_0) \prod_{k=0}^{n-1} \frac{G_k(X_k)}{H_k(X_k)}.$$

$$\underbrace{G_k - H_k}_{\text{Bellman defect}} = 0 \quad \text{for exact values.}$$

Approximate values leave a path defect. A direct continuous residual uses time derivatives and a Hessian trace.

Can function values supply both guidance and the missing correction?

Source: Stein guidance manuscript, path factorization for approximate Doob values.

Stein links value selection to gradient guidance

GAUSSIAN INTEGRATION BY PARTS

Stein identity

$$\mathbb{E}[Zf(Z)] = \mathbb{E}[\nabla f(Z)], \quad Z \sim N(0, I).$$

Set $Y = \mu + \sqrt{h}\sigma Z$, $f(Z) = H(Y)$:

$$\mathbb{E}[ZH(Y)] = \sqrt{h}\,\sigma^\top \mathbb{E}[\nabla H(Y)].$$

Random direction $\times$ value

reveals an average slope.

EXACT LOCAL REJECTION, WHEN AN ENVELOPE EXISTS

Propose Y from the base Gaussian. Accept with probability $H(Y)/M$, where $0 < H \le M$.

$$G_h(\mu) = \mathbb{E}H(\mu + \sqrt{h}\sigma Z).$$

The accepted mean

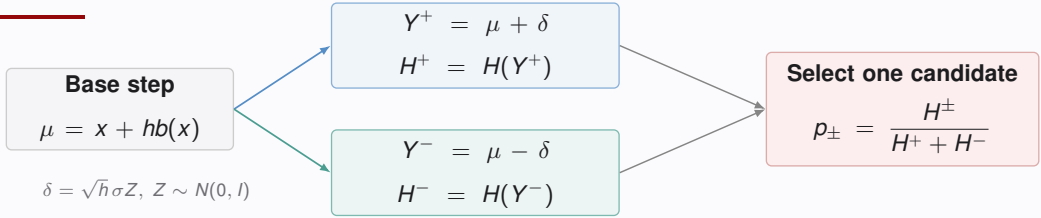
$$\mathbb{E}[Y \mid \text{accept}] = \mu + h\sigma\sigma^\top \nabla_\mu \log G_h(\mu).$$

The exact identity uses the smoothed value G_h. It matches a mean, not a shifted Gaussian law.

Finite pools replace these expectations with parallel value queries.

Source: Lu (2026), blog, Sec. 7; Zhu et al. (2026), Eq. (6), Eq. (23), and Proposition 6.

Two symmetric queries choose a direction



The value contrast gives the leading Doob drift

$$\mathbb{E}[Y^J - \mu \mid Z] = \delta \frac{H^+ - H^-}{H^+ + H^-} \approx \delta\delta^\top \nabla \log H(\mu).$$

$$\mathbb{E}[Y^J \mid x] = \mu + h\sigma\sigma^\top \nabla \log H(\mu) + o(h), \qquad S = \frac{H^+ + H^-}{2}.$$

Positive smooth H. Keep S for the path correction.

Source: Lu (2026), blog, Sec. 6; Zhu et al. (2026), Sec. 3.1.1, Eq. (5).

A centered pool extends two queries to any K

A BALANCED SET OF RANDOM DIRECTIONS

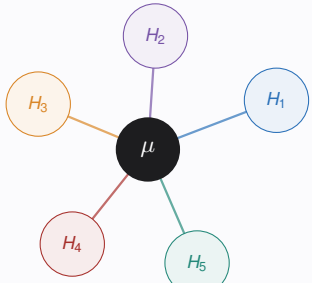


Illustration: $K = 5$, not five independent directions

Each $Z_i \sim N(0, I)$; pool directions are dependent.

Centering cancels the $\sqrt{h}$ term in the pool average.

DRAW, CENTER, AND RESTORE MARGINAL VARIANCE

$$G_i \overset{\text{iid}}{\sim} N(0, I), \qquad \bar{G} = \frac{1}{K} \sum_i G_i,$$

$$Z_i = \sqrt{\frac{K}{K-1}} (G_i - \bar{G}), \qquad \boxed{\sum_i Z_i = 0}.$$

Query in parallel, then select

$$Y_i = \mu + \sqrt{h}\sigma Z_i, \qquad H_i = H(Y_i),$$

$$\Pr(J = i \mid U) = \frac{H_i}{\sum_j H_j}, \qquad S = \frac{1}{K} \sum_i H_i.$$

Source: Zhu et al. (2026), Sec. 3.1.2, Eqs. (7) and (8); Lu (2026), blog, Sec. 7.

The same queries provide the path weight

AT EACH SOLVER STEP

One pool, two outputs

$$S_k = \frac{1}{K}\sum_{i=1}^{K} H_{k+1}(Y_{k,i})$$

Choose X_{k+1} with probability proportional to its value.

Keep both S_k and $H_{k+1}(X_{k+1})$.

No extra value query. No value derivative.

ACCUMULATE THE PROBABILITY COMPENSATION

Weight for a complete proposal

$$R = \phi(X_n)\prod_{k=0}^{n-1}\frac{S_k}{H_{k+1}(X_{k+1})}.$$

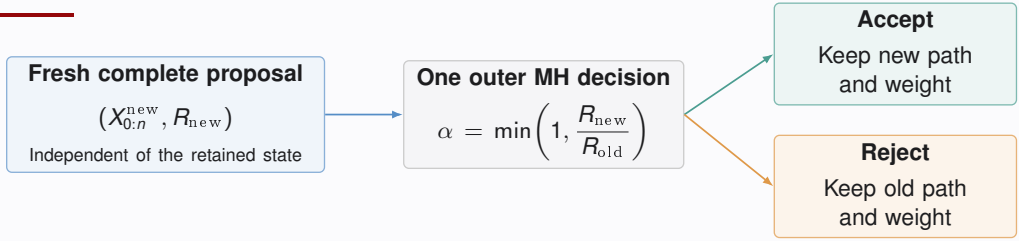
$$\log R = \log \phi(X_n) + \sum_{k=0}^{n-1} \log \frac{S_k}{H_{k+1}(X_{k+1})}.$$

Valid on the augmented space of pools and choices, not by an arbitrary density substitution.

Relative values guide. The same pool average corrects.

Source: Lu (2026), blog, Sec. 8; Zhu et al. (2026), Algorithm 1 and Theorem 12.

Path MH corrects after generation



What is independent?

Complete proposals can be generated independently and batched in parallel.

What is not independent?

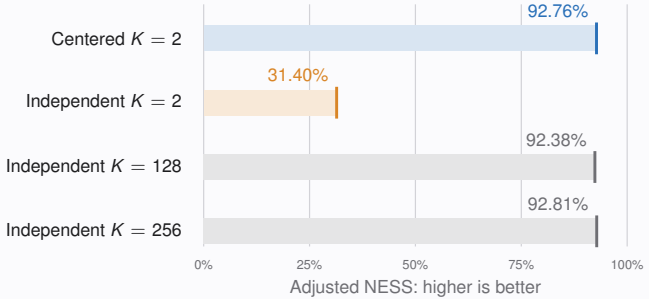
Consecutive outputs of one MH chain may repeat and remain correlated.

The target is invariant; a finite MH run can still have initialization error.

Source: Lu (2026), blog, Sec. 8; Zhu et al. (2026), Algorithm 2.

Two centered candidates stabilize weights

BROWNIAN BENCHMARK: PERTURBED VALUE, $n = 192$



Two queries suffice in this test

92.76%

Independent pools require 128 or 256 candidates to reach a similar level in this test.

Adjusted NESS evaluates normalizer estimation. It is not image quality or the number of independent MH outputs.

Source: Zhu et al. (2026), Sec. 5.1 and Figure 2; Lu (2026), blog, Sec. 10. Values redrawn from the reported results.

Path correction improves the target match

Hopper: fixed diffusion policy and critic, averaged over eight fixed states.

CONDITIONAL CRITIC SCORE KS DISTANCE: LOWER IS BETTER

Target mass	Guided only	With path MH
30%	0.2174	**0.0306**
60%	0.2804	**0.0697**
80%	0.4041	**0.1128**

Target mass is placed on the original policy's top 10% of actions by critic score.

Same two candidate guidance

$K = 2$ at every diffusion step.

Correction uses 32 complete proposals:
1 initialization + 31 outer updates.

What the experiment measures

Critic score distribution matching, not realized return.

A useful direction does not remove the need for probability correction.

Source: Zhu et al. (2026), Sec. 5.2, Table 2, Appendix H.3; Lu (2026), blog, Sec. 10.

One correction principle across the applications

Setting	Learned object	Defect	Inference computation
Integration	$\widehat{f}$	$f - \widehat{f}$	Residual Monte Carlo
Scientific ML	$\widehat{u}$	Bellman residual	Feynman Kac or MLP
Diffusion	Value H	Path probability mismatch	Stein selection and path MH

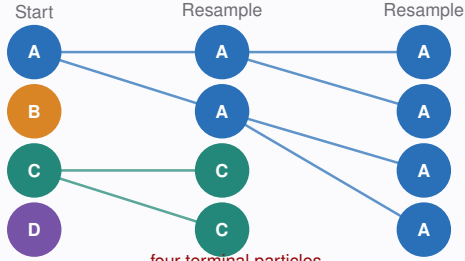
Diffusion sampling

Repeated resampling can create shared ancestry. Centered value queries provide a guidance direction and its matching correction weight.

Numerical refinement and independent sampling spend compute in different ways.

Shared ancestry creates covariance

REPEATED RESAMPLING COPIES ENTIRE PREFIXES



four terminal particles
one surviving stochastic family

WHY LOCAL DIAGNOSTICS MISS IT

Everything can look healthy

Flat weights can give ESS $= N$; even every single particle marginal can be correct.

But the empirical mean sees pair covariance

$$\text{For } \bar{f}_N = N^{-1} \sum_i f(X_n^i),$$

$$\mathrm{Var}(\bar{f}_N) = \frac{1}{N} \mathrm{Var}(f(X_n^1)) + \frac{N-1}{N} \mathrm{Cov}(f(X_n^1), f(X_n^2)).$$

***N* endpoints do not mean *N* independent histories.**

Source: Supplied manuscript, neutral Brownian SMC construction and variance decomposition.

The derivative cost of a direct correction

GUIDANCE USES A SPATIAL GRADIENT

$$dX_t = (b_t + a_t \nabla \log H_t)\, dt + \sigma_t\, dB_t.$$

A DIRECT BELLMAN RESIDUAL USES CURVATURE AND TIME VARIATION

Continuous residual

$$\mathcal{E}_t = \frac{\partial_t H_t + b_t \cdot \nabla H_t + \frac{1}{2} a_t : \nabla^2 H_t}{H_t}, \qquad R_H = H_0(X_0) \exp\left(\int_0^T \mathcal{E}_t(X_t)\, dt \right).$$

Time derivatives and Hessian traces can be costly for a learned value model.

Can function queries provide both guidance and correction?

Source: Stein guidance manuscript, continuous Bellman residual formulation.

Centering cancels the leading pool noise

THE SAME POOL AVERAGE WILL ENTER THE PATH WEIGHT

$$S = \frac{1}{K}\sum_i H(\mu + \sqrt{h}\sigma Z_i) = H(\mu) + \underbrace{\sqrt{h}\, g^\top \bar{Z}}_{\text{linear random term}} + O_{L^2}(h), \qquad g = \sigma^\top \nabla H(\mu).$$

Independent candidates

$\mathbb{E}\bar{Z} = 0$, but $\bar{Z}$ varies from pool to pool.

$\sqrt{h}$ scale fluctuations

The linear term vanishes only after averaging.

Centered candidates

$\bar{Z} = 0$ for every pool.

h scale fluctuations

The linear term cancels before averaging.

Centering removes the linear noise term using the same query budget.

Small step comparison with fixed K, positive smooth values, and controlled second order remainder. The smoothing bias is a separate issue.

Source: Zhu et al. (2026), Remark 2; Lu (2026), blog, Sec. 7.

Why the direction is a log gradient

SUBTRACT: RECOVER A DIRECTIONAL SLOPE

Odd part

$$H^+ - H^- \approx 2\delta^\top \nabla H(\mu).$$

ADD: NORMALIZE BY THE VALUE

Even part

$$H^+ + H^- \approx 2H(\mu).$$

SELECTION COMBINES BOTH OPERATIONS

$$\frac{H^+ - H^-}{H^+ + H^-} \approx \delta^\top \nabla \log H(\mu).$$

Then average over the random direction:

$$\mathbb{E}[\delta\delta^\top] = h\sigma\sigma^\top.$$

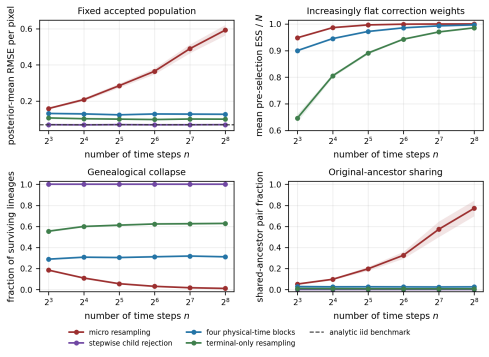
Two value queries recover the leading Doob drift

$$\mathbb{E}[Y^J \mid x] = \mu + h\sigma\sigma^\top \nabla \log H(\mu) + o(h).$$

This leading order mean identity assumes positive smooth H. It does not identify the full transition law.

Source: Zhu et al. (2026), Sec. 3.1.1, Eq. (5); Lu (2026), blog, Sec. 6.

Rejection stays stable as SMC genealogy collapses



CONTROLLED LINEAR IMAGING

$N = 160$ accepted paths · 36 runs

$n = 8, 16, 32, 64, 128, 256$

Sequential rejection

$$\mathrm{RMSE} : 0.0683 \text{ to } 0.0696$$

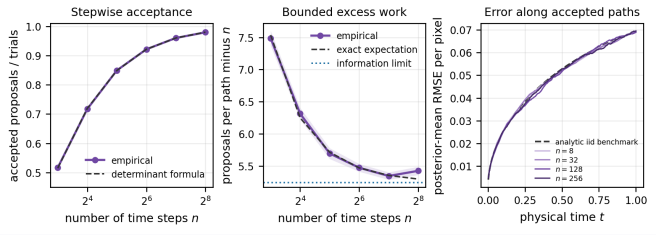
Full resampling SMC

$$\mathrm{RMSE} : 0.159 \text{ to } 0.593$$

At the finest mesh, ESS is $0.99994N$, yet only 1.02% of lineages survive and 77.2% of terminal pairs share an ancestor.

Rejection stays on the iid benchmark while repeated SMC cloning collapses genealogy.

Rejection trades shared ancestry for local retries



$$W_n = \sum_{k=1}^{n} \alpha_{k,n}^{-1} = n + I(X_1; Y) + O(n^{-1})$$

Acceptance: $0.517 \to 0.979$

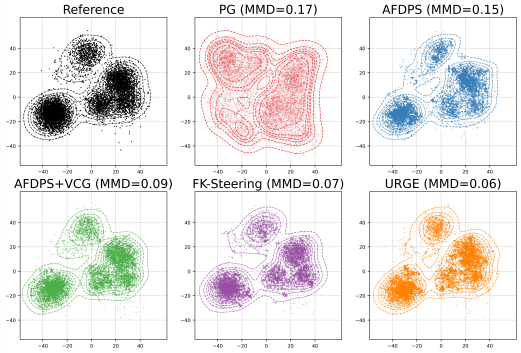
Excess proposals: $7.54 \to 5.30$

Information limit: $I(X_1; Y) = 5.239$ nats

Information controls retry overhead; population controls independent sampling variance.

Source: Supplied diffusion imaging manuscript, rejection diagnostics and exact Gaussian work formula.

Pathwise correction recovers the target modes



On the reported 30 dimensional Gaussian mixture, URGE gives the closest sample cloud to the analytical target, with **MMD** 0.06.